

Illustris Simulation Model (Mark Vogelsberger)

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Agha Tasheer Syedi¹

cs.aghasyedi@gmail.com

¹School of Quantum Technology, DIAT, Pune, India

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THE COSMIC SANDBOX: PLAYING GOD WITH A COMPUTER

The universe is not merely a collection of isolated, glowing objects scattered across an expanding void; it is a profound, interwoven tapestry of matter, energy, and forces acting over unimaginable timescales. When astrophysicists gaze into the deep sky, they see the final frame of a film that has been running for nearly fourteen billion years. But seeing the final frame does not tell you the plot. To truly understand the universe, we must rewind the tape. We must find the script, cast the actors, and hit 'play' to see if the story unfolds exactly as we observe it. This chapter explores the conceptual foundation of the Illustris Simulation, an audacious attempt to regrow a universe from scratch inside a supercomputer. We are not just building models; we are constructing a grand cosmic sandbox where the fundamental laws of nature are the only rules, and the entire architecture of the cosmos is the inevitable consequence. Let me tell you what this really means.

Building a Universe in a Box

Imagine you are standing in a bustling Indian kitchen on a hot summer afternoon, watching your grandmother prepare a complex, slow-cooked biryani. She doesn't have a modern recipe book; she has intuition, a deep understanding of how the spices interact, how the heat permeates the rice, and how the steam must be trapped to coax out the perfect flavours. If you understand the ingredients—the basmati rice, the cardamom, the saffron—and you understand the rules of the kitchen—the precise application of heat, the exact timing—you can predict, to a reasonable degree of accuracy, what the final dish will taste like.

The Illustris Simulation is much like that biryani, but on a scale that defies ordinary comprehension. The ingredients are not rice and spices, but the fundamental constituents of reality: dark matter, which acts as the unseen scaffolding; normal matter, the hydrogen and helium gas that will eventually form stars; and a subtle, omnipresent energy driving the expansion of space itself. The 'rules of the kitchen' are the unbreakable laws

of physics: gravity, thermodynamics, and the bizarre behaviour of fluids under immense pressure.

We take these ingredients, we define the starting conditions—the initial temperature and density of the 'oven' immediately following the Big Bang—and we let the computer simmer this cosmic stew for a simulated 13.8 billion years. We are not just observing the universe; we are actively attempting to bake our own, within a digital box measuring hundreds of millions of light-years across. If our recipe is correct, and if our understanding of the fundamental forces is true, the resulting digital universe should look suspiciously like the one we see through our telescopes. If it doesn't, we know our understanding of the recipe is flawed.

The Grand Misconception: It is Not Just a Movie

When you first see a visualisation of the Illustris Simulation, it is a profoundly moving experience. You watch wispy filaments of gas coalesce into glowing galactic hubs, spiralling beautifully in the dark void. It is easy, almost natural, to assume you are watching an incredibly high-budget special effects sequence, akin to a Bollywood science-fiction epic where every star and galaxy has been carefully placed by an artist for maximum dramatic effect.

This is the grand misconception we must shatter immediately. In a CGI film, a visual artist draws the shape of a spiral galaxy. In Illustris, no human hand ever draws a galaxy. The computer has absolutely no idea what a 'galaxy' is supposed to look like. The programmers did not write a line of code that said, 'At year 5 billion, create a beautiful blue spiral here.'

Instead, the computer is only taught the blind, unthinking laws of physics. It is taught how gravity pulls masses together, how gas heats up when it is compressed, and how radiation from young stars pushes back against the surrounding material. The breathtaking spiral arms, the blazing golden cores, the delicate lattice of the cosmic web—these are not artistic choices. They are emergent phenomena. They are the inevitable, raw, unedited consequences of the fundamental laws of nature playing out over deep time. When we see a beautiful digital galaxy in Illustris, we are not

looking at art; we are looking at the stark, brutal honesty of physics revealing its inherent beauty.

A Brief History of Cosmic Cartography

To appreciate the leap forward that Illustris represents, we must understand the history of our cosmic map-making. For many years, theoretical physicists tried to simulate the universe using only gravity and the elusive substance we call 'dark matter'. These early models, often called N-body simulations, were ground-breaking, but they were deeply simplified.

Imagine trying to model the chaotic, vibrant energy of a local bazaar on the eve of Diwali, but instead of modelling individual people with their complex desires and interactions, you model everyone as a perfectly smooth, identical, frictionless sphere. You might capture the general flow of the crowd, the way people clump around popular stalls, but you would miss the essence of the bazaar—the noise, the haggling, the sudden bursts of movement.

These early simulations treated the universe as a swarm of perfectly smooth, frictionless marbles representing dark matter. They successfully produced the invisible, overarching structure of the universe—the vast, sprawling filaments of dark matter known as the cosmic web. But this universe was entirely dark. You cannot build a glowing, turbulent star out of a frictionless marble. To create the vibrant, luminous cosmos we inhabit—to get stars, planets, and the potential for life—astrophysicists realised they had to abandon the clean, mathematical perfection of pure gravity and embrace the messy, chaotic, sticky friction of normal, everyday matter.

The Fluids of the Cosmos

This brings us to the realm of hydrodynamics—the physics of fluids in motion. Think about stirring milk into a cup of hot chai on a rainy evening. Watch carefully how the fluid behaves. The milk doesn't just instantly dissolve; it swirls, it creates complex eddies, it cools down as it interacts with the air, and it forms intricate, temporary structures before settling.

In the grand arena of the cosmos, the vast clouds of hydrogen and helium gas behave exactly like a fluid. As gravity relentlessly pulls this cosmic gas into the immense gravitational 'valleys' created by the dark matter scaffolding, the gas streams crash into one another. The kinetic energy of their motion is violently converted into heat, raising the temperature of the gas to millions of degrees.

It is this turbulent fluid dynamics—the very same physics that dictates the swirling patterns of the monsoon clouds rolling over the Western Ghats—that ultimately sculpts the grandest structures in the universe. The gas must cool, condense, and swirl into rotating discs before it can fragment and ignite into stars. The breathtaking spiral galaxies we see in the night sky are not solid objects; they are monumental whirlpools of cosmic fluid, governed by the same ancient laws that stir your evening tea.

THE RULES OF THE GAME: CODING THE LAWS OF NATURE

To build a universe in a computer, one cannot simply ask the machine to 'make galaxies'. A computer possesses no imagination; it only understands logic, mathematics, and strict rules. Therefore, the task of the computational astrophysicist is not to design the cosmos, but to encode the fundamental laws of nature—the ultimate rules of the game—and allow the universe to build itself. This section delves into the transition from the relatively simple physics of invisible scaffolds to the staggering complexity of fluid dynamics, star birth, and cosmic regulation. We are transitioning from the silent, ghostly dance of gravity to the violent, fiery physics of the visible universe. Let us look at the code of reality.

Gravity and the Dark Scaffold

Before a single virtual star can ignite, before the first drop of light pierces the digital darkness, the simulation must construct the unseen architecture of the cosmos. Think of it like building a magnificent temple in ancient India. Long before the intricate carvings of deities and lotus flowers are added to the facade, a massive, unseen foundation of heavy stones must be laid deep into the earth. The visible temple rests entirely upon this invisible strength.

In our universe, this foundation is made of dark matter. Dark matter is a profound mystery, but for the purposes of a simulation, it is beautifully simple: it only interacts through gravity. It does not feel friction, it does not emit light, and it does not bounce when it collides.

The computer begins by distributing billions of virtual 'particles' of dark matter. It then calculates the gravitational pull that every single particle exerts on every other particle, using the classical laws discovered by Isaac Newton, with necessary adjustments for the overarching expansion of space described by Albert Einstein. Imagine the sheer computational terror of this: if particle A pulls on particle B, the computer must also calculate how B pulls on A, and how both pull on C, D, and E, billions of times over, updating their positions frame by frame.

Over billions of simulated years, this ghostly matter begins to clump. Gravity is relentlessly patient. Minor ripples in the initial distribution of matter slowly amplify. The particles slide into deep gravitational 'valleys', forming massive, dense structures we call 'halos'. These dark matter halos are the cosmic cribs—the foundational stones—where galaxies will eventually be born. Visible galaxies, with all their swirling stars and glowing gas, are merely the luminous icing resting on top of massive, invisible mountains of dark matter.

The Hydrodynamical Leap

If a universe consisted only of dark matter, it would be a cold, silent, and completely invisible place. To add the 'normal' matter—the baryons that make up you, me, the Earth, and the stars—the Illustris team

had to confront hydrodynamics. They had to teach the computer how fluids behave on a cosmic scale.

This is a monumental leap in complexity. Governing fluid flow requires solving notoriously difficult equations that describe how pressure, density, temperature, and velocity interact in a continuous medium. Imagine standing on the platform of the Chhatrapati Shivaji Maharaj Terminus in Mumbai during the evening rush hour. People (like gas particles) flow through the station. When two massive crowds moving in opposite directions collide, they don't simply pass through one another like ghosts; they compress, they slow down, tempers flare, and 'heat' is generated.

When streams of cosmic gas fall into the gravitational grip of a dark matter halo, they accelerate to phenomenal speeds. When these streams crash into each other, they cannot simply pass through. They slam together, creating colossal cosmic shockwaves that convert the kinetic energy of their movement into raw thermal heat, raising the temperature of the gas to millions of degrees.

Herein lies the central drama of galaxy formation: a battle of thermodynamics. Gravity is relentlessly trying to pull the gas inward, demanding collapse. But the extreme heat creates thermal pressure, an outward force that furiously resists the collapse. For a galaxy to form, for the gas to become dense enough to forge stars, the gas *must* lose this heat. It must cool down. The simulation must continuously, meticulously calculate how this gas radiates its heat away into the expanding vacuum, balancing the inward crush of gravity against the outward push of thermal pressure.

Cooling, Star Formation, and Cosmic Feedback

If gravity always won this battle without resistance, the story of the universe would have been short and deeply uninteresting. All the gas would have collapsed into supermassive stars almost immediately after the Big Bang, burned through its nuclear fuel in a cosmic eye-blink, and left behind a dead, dark universe billions of years ago.

Why didn't this happen? The answer lies in a concept called 'feedback'.

Imagine you are trying to boil a pot of milk. You turn the stove up high. The milk gets hotter and hotter, rising in the pot, threatening to boil over in a chaotic mess. Just before it spills, you turn down the heat. The milk settles. You turn the heat back up, it rises again. You are regulating the boil through feedback.

Galaxies do exactly the same thing. When the simulated gas in Illustris gets sufficiently cold and incredibly dense, the computer spawns a 'star particle'—a proxy representing a massive cluster of new-born stars. But massive stars are the rock stars of the cosmos: they live fast, burn brilliantly, and die violently.

Within a few million years, these massive stars explode as supernovae. These explosions are unimaginably powerful, blasting immense amounts of energy and newly forged heavy metals back into the surrounding gas. This violent injection of energy heats the surrounding gas dramatically, physically blowing it outward and

temporarily shutting down any further star formation. The galaxy has boiled up, and the supernovae have effectively turned down the heat. The galaxy regulates its own growth through this violent, cyclical breathing—a rhythm of stellar birth and explosive death—preventing itself from consuming all its available fuel at once.

The Fine-Tuning Problem in Code

A simulation, no matter how powerful, is ultimately only as good as the numbers you feed into it. The Illustris model is built upon our current best understanding of the universe, a framework that relies on specific cosmological parameters. We must tell the computer the total density of matter, the amount of dark energy driving the universe apart, and the exact rate at which space itself is expanding.

But it goes deeper than that. The programmers must also mathematically define the strength of the physical processes, like the 'feedback' from those exploding stars. And this is where the simulation reveals a profound philosophical truth about our reality.

If the programmers tweak the virtual knobs even slightly—if they make the supernova explosions just a fraction of a percent stronger or weaker—the resulting universe looks entirely wrong. If the explosions are too weak, the gas never stops collapsing, and the simulation produces galaxies that are impossibly massive, dense blobs of stars unlike anything we see in reality. If the explosions are too strong, the gas is blown away so violently that galaxies never manage to form at all; the universe remains a diffuse, barren wasteland.

By tuning the parameters of this digital universe until the output finally matches the real galaxies we observe in the night sky, we inadvertently discover the precarious nature of our existence. We see, laid bare in the code, just how delicately balanced the physics of our reality truly is. A slight nudge to the underlying math, a minor alteration to the rules of the game, and the universe as we know it—capable of supporting stars, planets, and observers to wonder at it all—would simply cease to exist.

THE DEEP CODE: NAVIER-STOKES MEETS THE MULTIVERSE

As we move deeper into the architecture of the Illustris Simulation, we leave behind the intuitive, macroscopic physics of swirling gases and colliding clouds. We must now confront the formidable mathematical machinery required to force a computer to comprehend the infinite and the infinitesimal simultaneously. How do you map a continuous, flowing reality onto the discrete, rigid logic of binary code? How do you track the birth of a galaxy when the most violent physical processes happen on scales far smaller than your smallest measuring stick? This section explores the elegant mathematical geometry that makes the simulation possible, the necessary compromises physicists must make at the quantum limit, and the profound way information theory reframes our understanding of a galaxy.

Discretising the Infinite: Moving-Mesh Mathematics

One of the greatest challenges in computational physics is how to represent a continuous fluid—like water, air, or cosmic gas—inside a machine that only understands discrete numbers.

Historically, physicists had two primary ways of tackling this. The first, the Eulerian approach, is like placing a rigid wire grid over a flowing river. You don't track the water itself; you just record how fast the water is moving and how dense it is as it flows *through* each fixed square of your grid. The problem? If the river suddenly shrinks into a narrow, violent rapid, your rigid grid might be too coarse to capture the intricate details of the turbulence.

The second method, the Lagrangian approach, is like throwing millions of tiny, floating leaves into the river and tracking the path of every single leaf. This captures the flow beautifully, but if a whirlpool forms and violently stretches a group of leaves apart, you lose detail in the empty spaces between them.

The creators of Illustris needed something better, a way to capture the sheer, dynamic scale of the cosmos—from the vast, empty voids to the unbelievably dense cores of galaxies. They used a revolutionary code called AREPO, which employs a mathematical concept known as a Voronoi tessellation.

Imagine you are looking at the cracked mud of a dried Indian riverbed during the peak of summer. The cracks form a chaotic, interlocking network of polygons. Now imagine that this network is three-dimensional and that the polygons can move, stretch, shrink, and flow.

This is the genius of the AREPO 'moving mesh'. Instead of a rigid grid, the computer creates a flexible mesh of 3D polygons that actually drift *with* the flow of the cosmic gas. Where the gas is thick and dense—like gas collapsing to form the brilliant core of a new spiral galaxy—the mesh cells automatically shrink, clustering tightly together to provide incredibly high-resolution detail exactly where the physics is most complex. Where the gas is sparse, out in the lonely cosmic voids, the mesh cells expand, saving computational power.

We are no longer forcing the fluid into a rigid box; we have invented a geometry that breathes, flexes, and dances in perfect tandem with the matter it contains.

Sub-grid Physics and the Quantum Limit

Despite the elegance of the moving mesh, we immediately crash into a brutal reality: the sheer, uncompromising scale of the universe. Even with the world's most powerful supercomputers working in parallel for months, we cannot track an entire universe down to the scale of an individual atom, or even an individual star.

In the Illustris Simulation, a single computational 'cell'—one of those flexible polygons in the mesh—might represent a volume of space hundreds of light-years across.

Yet, some of the most violent and consequential events in a galaxy happen on scales vastly smaller than this. Consider the supermassive black hole lurking at the centre of a galaxy. The accretion disc—the swirling maelstrom of superheated gas screaming into the black hole—is tiny compared to the galaxy itself; it might be smaller than our own solar system. But the energy released by this tiny disc can blast powerful jets across millions of light-years, profoundly altering the shape of the entire galaxy.

How can the computer simulate an explosion driven by a black hole when the entire black hole, and its surrounding gas, is smaller than the computer's smallest pixel?

Physicists bridge this gap using 'sub-grid models'. These are essentially mathematical heuristics—educated physical approximations—placed inside the code. We tell the computer, "Look, if the gas in this massive cell gets this dense and this hot, don't try to simulate the individual atoms. Just assume a black hole is feeding, calculate the theoretical energy output based on our best quantum and relativistic models, and inject that raw energy back into the macroscopic grid."

The macro-architecture of the cosmos—the majestic spiral arms and the sprawling cosmic web—is inextricably chained to the violent, microscopic quantum physics happening below the resolution of our greatest machines. We cannot see the quantum engine, but we must mathematically inject its roar into the simulation for the galaxies to form correctly.

Observational Mimicry and Mock Catalogues

A theoretical physics model, no matter how mathematically beautiful or visually stunning, is entirely useless if it cannot be rigorously tested against reality. The Illustris team cannot just look at the code and declare it a success. They must prove it. But how do you compare a digital universe inside a hard drive to the real universe sitting above our heads?

You don't compare the code to the sky; you compare the *light*.

Scientists generate what are known as "mock observations." They write secondary programmes that act as virtual telescopes. They instruct the computer to trace the path of virtual photons (particles of light) as they are emitted from the simulated stars, travel through the simulated gas clouds, and navigate the expanding space-time of the digital universe.

They must account for everything. If a virtual galaxy is full of simulated dust, the computer must calculate exactly how much blue light that dust absorbs, making the galaxy appear redder—just like dust in the Earth's atmosphere turns the sun red at sunset. They even program the virtual telescope to have the exact same limitations, blurs, and blind spots as the real Hubble Space Telescope.

By building a virtual telescope inside the simulation, they create 'mock catalogues'—lists of digital galaxies with specific brightnesses, colours, and shapes. They can then place these mock catalogues side-by-side with

the real catalogues of galaxies observed by astronomers. It is an apples-to-apples comparison. The ultimate test of artificial reality is whether a virtual telescope, looking at a virtual universe, captures the exact same patterns of light as a real telescope looking at our own.

Information Theory in Galaxy Assembly

Let us step back and look at the simulation from a completely different, profoundly abstract perspective: information theory.

The early universe, shortly after the Big Bang, was a nearly uniform, featureless plasma. It was incredibly hot, incredibly dense, and almost perfectly smooth. In the language of thermodynamics, it was a state of high entropy—a state of maximum disorder, where every part looked exactly like every other part.

A galaxy, by contrast, is a masterpiece of order. It is a highly structured arrangement of stars, intricate chemical gradients, swirling dust lanes, and intricate planetary systems.

From an information theory perspective, the formation of a galaxy is a monumental, localized reduction of entropy. Gravity acts like a cosmic Maxwell's Demon—a theoretical entity that sorts chaos into order. Gravity relentlessly pulls the uniform gas together, sorting the matter into highly ordered, complex structures while radiating the 'waste heat' (the entropy) out into the expanding voids of space.

By tracking the exact 'metallicity' in the Illustris code—the precise distribution of heavy elements forged in the hearts of simulated supernovae—astrophysicists are not just tracking chemistry. They are quantifying the flow of thermodynamic information across cosmic time. A galaxy is not merely a collection of stars drifting in the dark; it is a localized engine of computation, converting the raw, chaotic potential of the early universe into the complex physical information necessary for chemistry, planets, and, eventually, life.

THE ULTIMATE FRONTIER: SIMULATING THE SIMULATORS

We have traversed the architecture of our digital cosmos, from the silent gravitational scaffolding to the fiery, turbulent birth of spiral galaxies. Yet, science is never truly finished; it merely arrives at more profound thresholds of ignorance. As our supercomputers grow exponentially more powerful, we are beginning to peer over the precipice of what it means to model reality. The original Illustris Simulation was a triumph, but it was not perfect. It produced galaxies that were occasionally too bloated, or star formation rates that did not perfectly align with our deepest telescopic observations. To bridge this gap, we must look beyond gravity and fluid dynamics. We are now entering an era where the philosophical implications of our work are as staggering as the physics itself. Let us explore the bleeding edge of the cosmic sandbox.

The Missing Physics: Magnetic Fields and Cosmic Rays

Imagine looking at a breathtaking, intricately woven Kanjeevaram silk sari from across a room. You see the grand sweep of the fabric, the bold colours, and the way it flows and drapes over a silhouette. This is how the original Illustris simulation viewed the universe: it captured the grand, sweeping fluid dynamics of gravity and gas. However, if you step close to the silk, you realise its shape and shimmering rigidity are governed by microscopic, nearly invisible threads of gold woven through the cotton.

In the cosmos, those invisible gold threads are magnetic fields. The original models largely ignored them, assuming they were too weak to matter on a galactic scale. We now realise this was a profound oversight. Galaxies are threaded with immense, invisible magnetic fields—a field of study known as magnetohydrodynamics. These fields act like an invisible nervous system, guiding the flow of gas, restricting turbulence, and shaping the grand spiral arms.

Furthermore, we must now account for cosmic rays. These are not rays of light, but ultra-high-energy protons zipping across the void at fractions of the speed of light, often accelerated by the shockwaves of dying stars. They do not behave like a normal fluid; they refuse to be tamed by standard thermodynamics. They punch through gas clouds like invisible bullets, exerting a subtle but relentless pressure that can literally push a galaxy apart from the inside. Capturing the kinetic impact of these ghostly particles requires mathematics so terrifyingly complex that it pushes modern supercomputers to their absolute breaking point. The deeper we model the universe, the more we realise that these whisper-thin 'minor details'—magnetism and invisible radiation—are the true, silent puppet masters of the visible cosmos.

Exascale Universes and the Resolution Limit

We are currently crossing the threshold into the era of Exascale computing, where machines can perform a quintillion calculations every single second. This raw power is driving us toward the holy grail of cosmological modelling: shattering the resolution limit.

Think of a satellite photograph of the Maha Kumbh Mela, the largest human gathering on Earth. From space, the tens of millions of pilgrims look like a continuous, swirling fluid of colours moving along the riverbanks. This is where our simulations are today. In the code, a single 'star particle' does not represent one star; it is a mathematical proxy for millions of stars lumped together. We are modelling the crowd, not the person.

But as we march relentlessly towards Zettascale computing in the coming decades, the resolution of our digital universes will sharpen dramatically. We will eventually cross a threshold where a simulation spanning hundreds of millions of light-years can 'zoom in' to resolve the birth of individual star systems. We will see the digital gas collapse not just into a galactic core, but into specific, rotating proto-planetary discs. We

are steadily, inevitably approaching the computational horizon where our simulated universes will possess the granular resolution necessary to host virtual Earths, complete with shifting tectonic plates and swirling digital atmospheres.

The Epistemological Crisis of Simulation

As we approach this god-like resolution, theoretical physics finds itself staring into the abyss of an epistemological crisis—a crisis of knowledge itself.

When you write a simple equation on a chalkboard, you understand every variable. But modern simulations consist of millions of lines of code, crunching numbers for months across tens of thousands of processors. The outputs are so overwhelmingly complex that no single human mind can comprehend them; we must now build Artificial Intelligence simply to read the data our simulations produce.

This creates a profound philosophical dilemma. When our AI discovers a strange, unexpected new structure in the digital cosmic web, how do we know if it is real? Are we actually discovering a new, fundamental truth about nature, or are we merely exploring the emergent, unintended consequences of the algorithms we ourselves invented? If the Voronoi tessellation of the moving-mesh math creates a subtle rounding error, magnified over thirteen billion simulated years, it might masquerade as a new law of physics. When the tools we use to understand reality become as impenetrably complex as reality itself, the boundary between discovering physics and hallucinating it begins to blur. We risk becoming lost in a labyrinth of our own making.

The Simulation Hypothesis Echo

This brings us to the ultimate, dizzying implication of our work. Humanity has possessed the formal mathematics of general relativity and quantum mechanics for barely a century. We have had programmable computers for less than a human lifetime. Yet, in this microscopic fraction of cosmic time, we have already successfully modelled the hydrodynamical formation of galaxies with stunning visual and physical fidelity.

Project this trajectory forward. What happens in a thousand years? What happens in ten thousand years? If an advanced civilisation survives its technological adolescence, it will inevitably possess computing power capable of simulating a universe with perfect, atomic resolution. And if they can run one simulation, they would likely run billions of them, perhaps to study their own history, or simply as cosmic art.

This is the chilling echo of philosopher Nick Bostrom's Simulation Hypothesis. It argues that if universes *can* be simulated, then the number of simulated universes will vastly outnumber the single 'base' reality. Statistically speaking, if this is true, the vast majority of conscious observers are inside a machine.

For decades, this was merely a fun philosophical thought experiment. But the Illustris Simulation changes the paradigm. It is the first crude, macroscopic proof of concept that creating a universe in a machine

is not just a science fiction trope, but a mathematically achievable reality. By successfully coaxing a digital cosmos to bloom inside our supercomputers, we have inadvertently provided the most compelling evidence to date that our own universe, with all its shimmering galaxies and deep mysteries, might just be someone else's simulation running on a higher-dimensional hard drive.